PROCESSES AND METHODS IN INTERACTION DESIGN

TEST

Redesigning Google Classroom

Creating a More Interactive & Inclusive Learning Platform

ILISHA DABAS 2k24/BD/046 SECTION-A

Introduction to Google Classroom

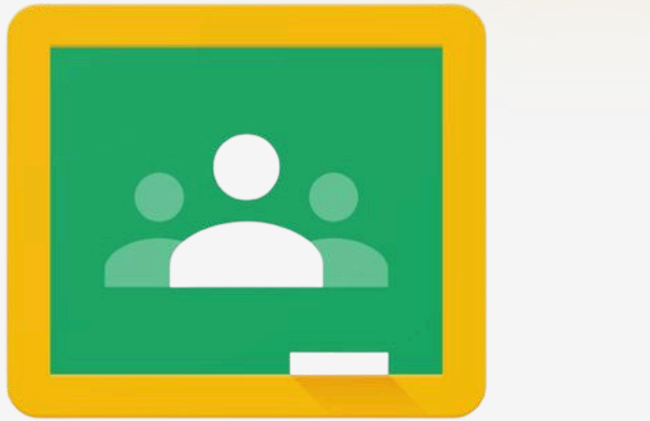
Google Classroom is a widely used Learning Management System (LMS) that helps teachers and students manage digital classrooms.

It allows teachers to:

- Share assignments
- Post announcements
- Organize class materials
- Communicate with students

While the platform is functional and widely adopted, the user experience is mostly focused on assignment distribution rather than active learning engagement.

This creates an opportunity to redesign the platform to improve interaction, collaboration, and accessibility.



Google Classroom

Problems in the Current Platform

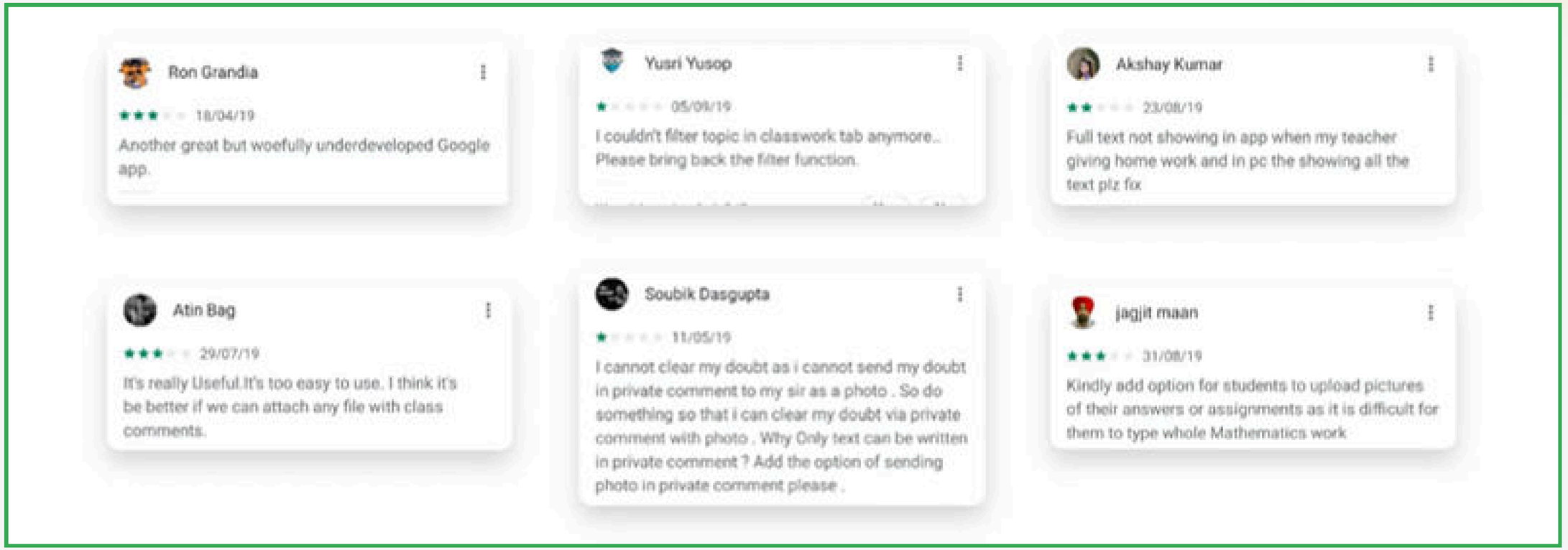


Despite its usefulness, the current platform has several limitations:

- Limited student interaction and participation
- Difficult to track assignments visually
- Minimal collaborative learning opportunities
- Lack of motivation or engagement features
- Interface focused mainly on task submission

These issues reduce student engagement and make the learning experience feel passive rather than interactive.

Problems in the Current Platform



Objective of the Redesign

Design Goal
The goal of this project is to redesign Google Classroom to create a more interactive, engaging, and inclusive digital learning environment.



The redesign focuses on:

- Improving student engagement
- Enhancing communication between students and teachers
- Providing better organization of assignments
- Supporting collaborative learning
- Creating a more intuitive and visually structured interface

Target Users

Students

- Need clear assignment organization
- Want engaging learning interactions
- Require easy access to resources and deadlines

The redesign aims to balance the needs of both user groups.

Teachers

- Need tools to manage classes efficiently
- Want insights into student performance
- Require easy communication channels with students



Research Insights

Key Observations
After analyzing existing LMS platforms and user needs, the following insights were identified:

Students prefer:

- Clear visual assignment tracking
- Interactive learning spaces
- Notifications and reminders

Teachers prefer:

- Simplified class management
- Better visibility of student progress
- Organized content distribution

These insights helped guide the design decisions.





Design Strategy Followed

Approach to Redesign

The redesign followed several UX principles:

- Simplicity and clarity
- Structured information hierarchy
- Visual progress indicators
- Interactive communication spaces
- Modular card-based layout

The goal was to create a **clean and intuitive interface that reduces cognitive load.**

Ideation

Brainstorming Solutions
Several ideas were explored during the ideation phase to improve engagement and usability.

Key concepts included:

- A dashboard showing assignment progress
- A collaborative class feed for discussions
- A centralized resource library
- A to-do panel for pending tasks
- Quick class joining system

These ideas aimed to create a more interactive and organized learning environment.



Key Features Implemented

The redesigned version of Google Classroom introduces features aimed at improving student engagement, organization, and classroom insights.

Gamification System

Introduces progress indicators, rewards, and achievements to motivate students and increase participation in class activities.

Interactive Student Dashboard

A centralized dashboard where students can view assignments, schedules, and important updates in a structured and easy-to-navigate layout.

Clear Assignment Categorization

Assignments are organized into categories such as upcoming, completed, and overdue, helping students prioritize tasks and manage deadlines effectively.

AI Learning Assistant

An integrated AI assistant that helps students understand concepts, clarify doubts, and get guidance while working on assignments.

Schedule & Planning

Students can track class schedules and upcoming deadlines, enabling better time management and planning.

Recent Activity Feed

Displays announcements, graded assignments, and class updates so students remain informed about the latest classroom activity.

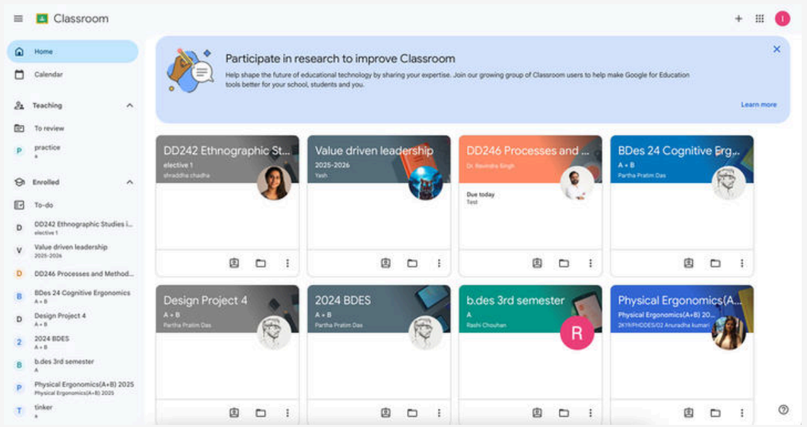
Teacher Activity Dashboard

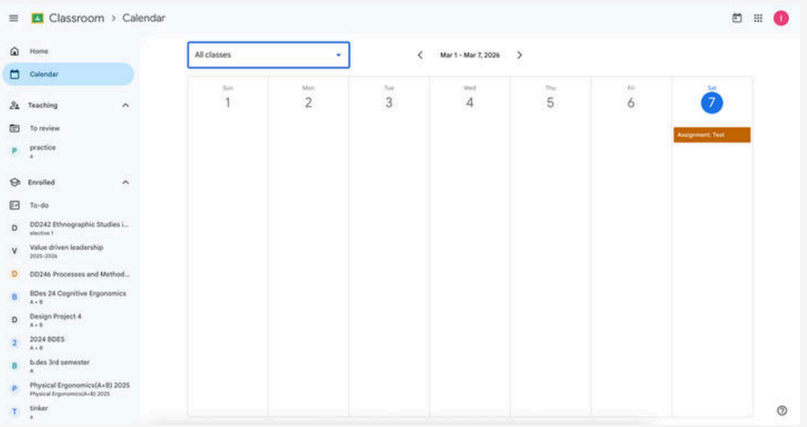
Provides teachers with insights into student engagement, participation, and performance, helping them identify students who may need additional support.

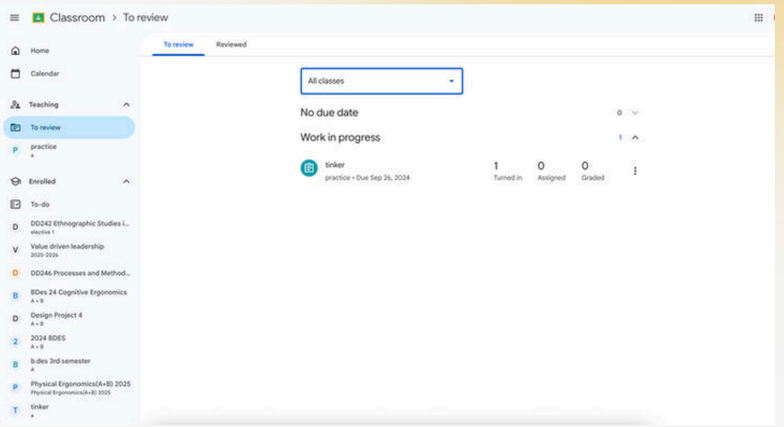
Impact:

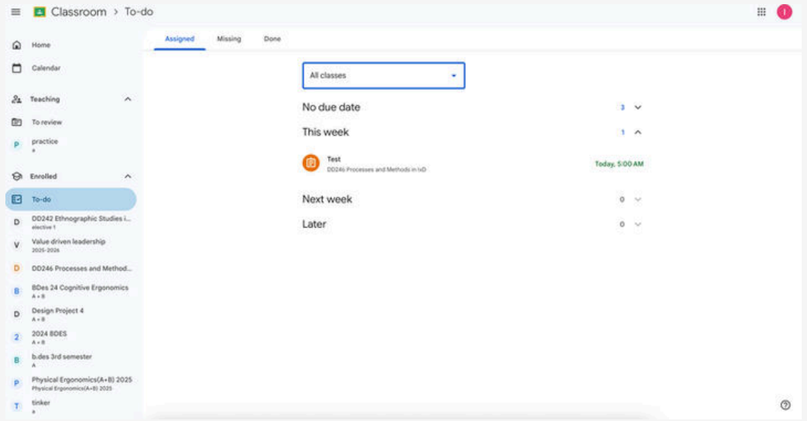
These features transform the platform into a more interactive, organized, and data-driven learning environment for both students and educators.

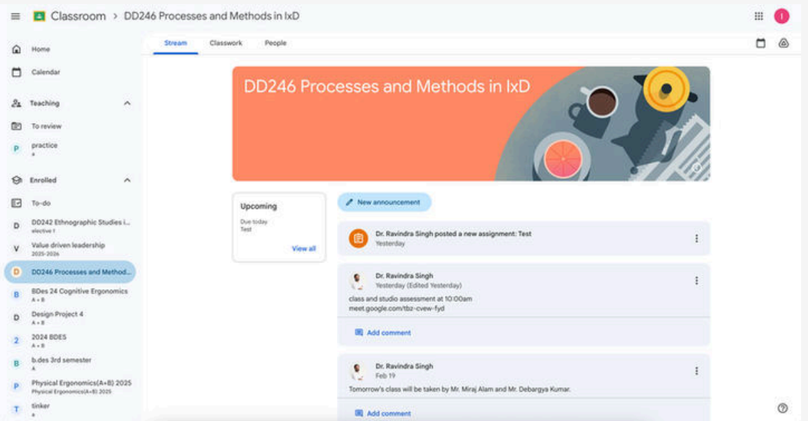
Original Desgin

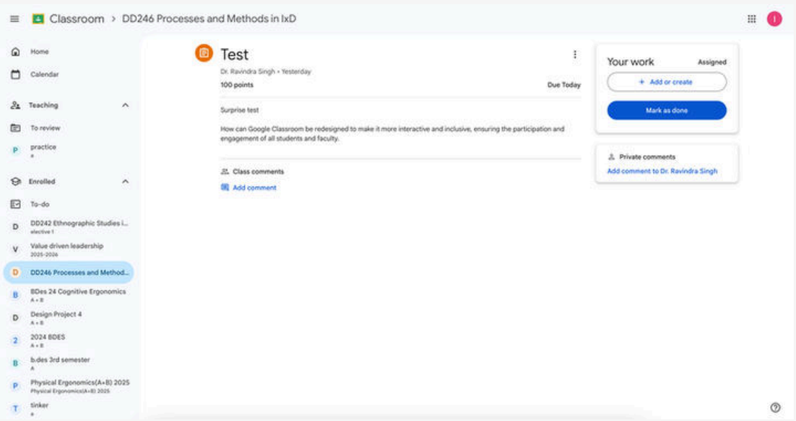




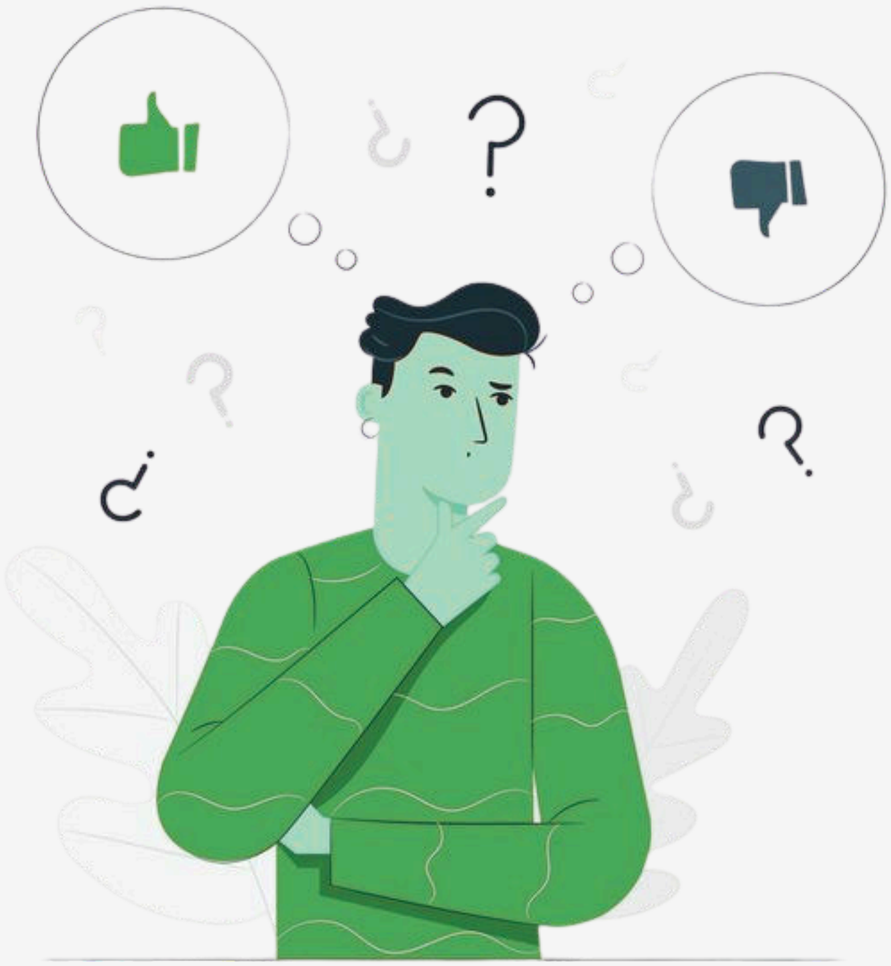








Inclusive Design Considerations



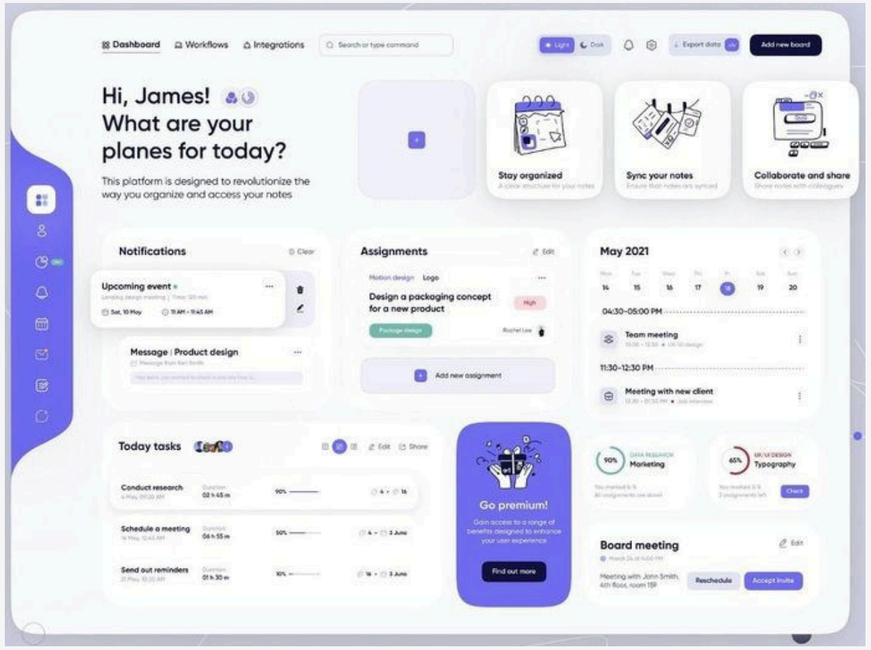
Designing for Accessibility

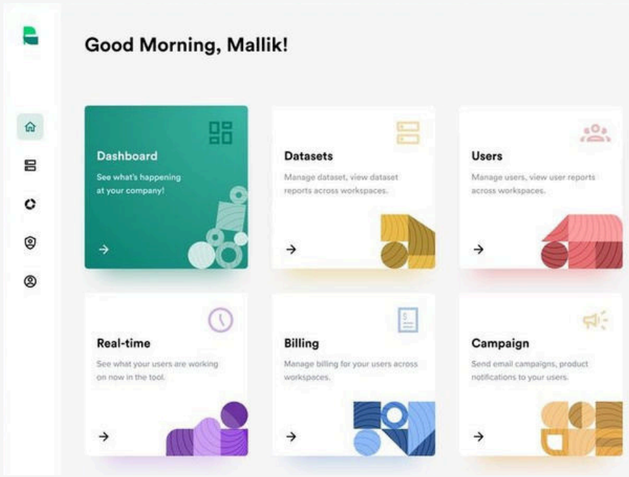
The redesign focuses on creating a more inclusive learning experience by:

- Using clear and readable typography
- Structuring content to reduce cognitive overload
- Providing visual indicators for assignments and deadlines
- Simplifying navigation for easier access to information

These elements help support diverse learning styles.

Redesign Moodboard

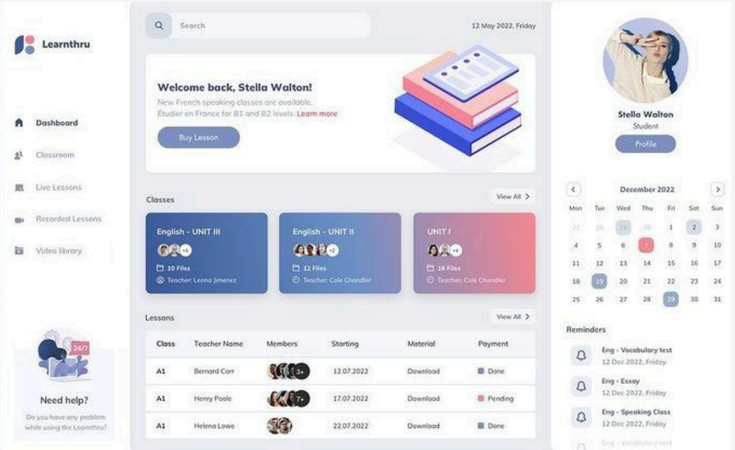




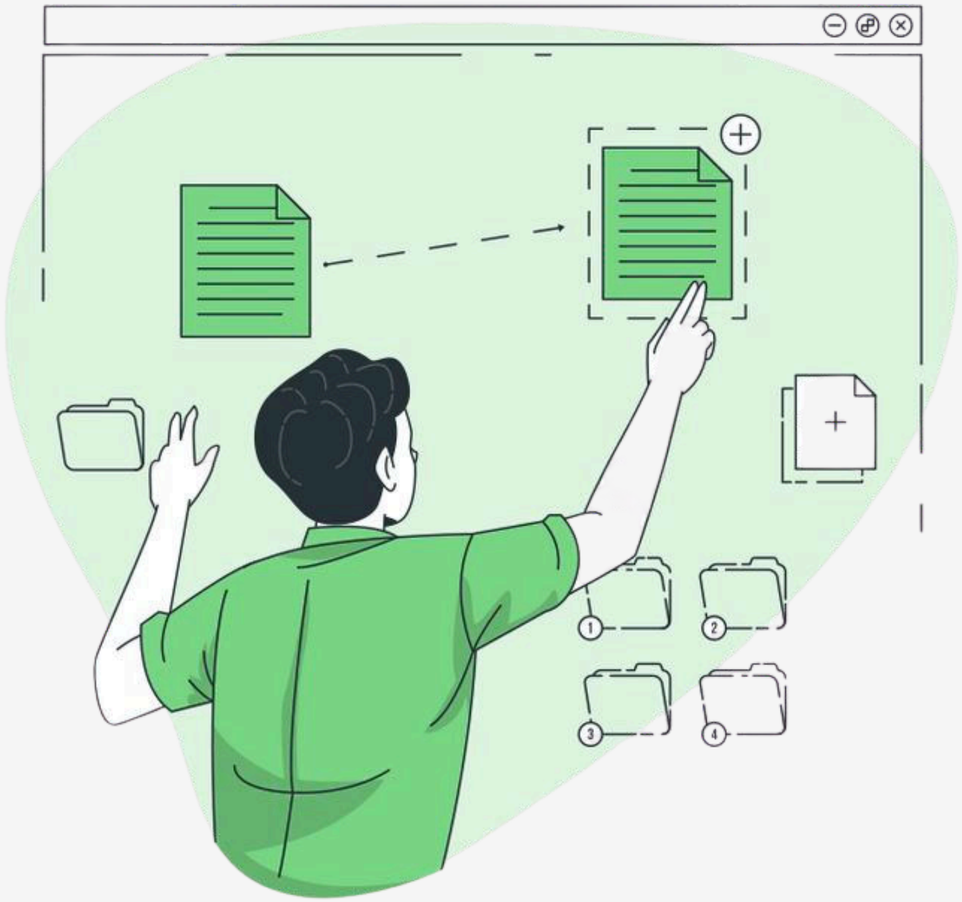








Design System



Visual Design Language

The interface follows a clean design system inspired by Google's design principles.

Color Palette

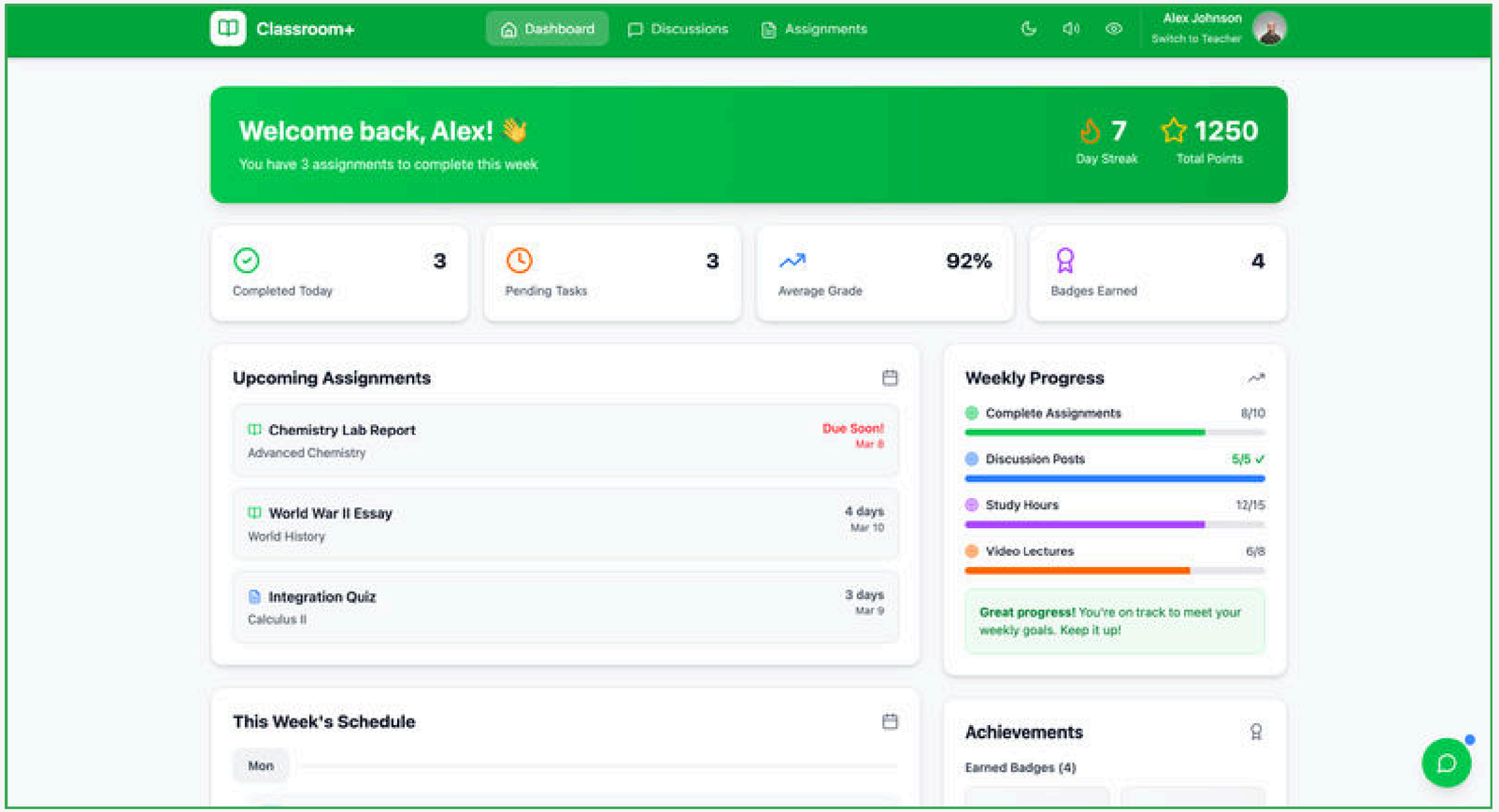
Primary Green: #34A853
Secondary Blue: #4285F4
Accent Yellow: #FBBC05

Design Elements

- Card-based layouts
- Minimal icons
- Clean typography
- Consistent spacing

This visual system ensures a modern and cohesive interface.

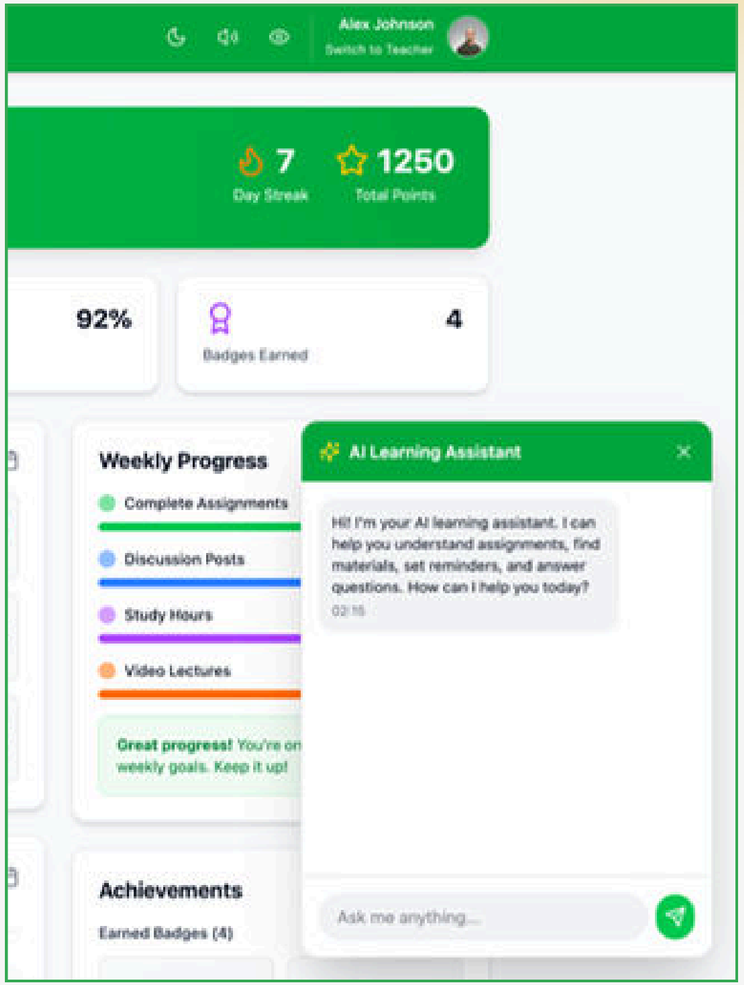
Redesign



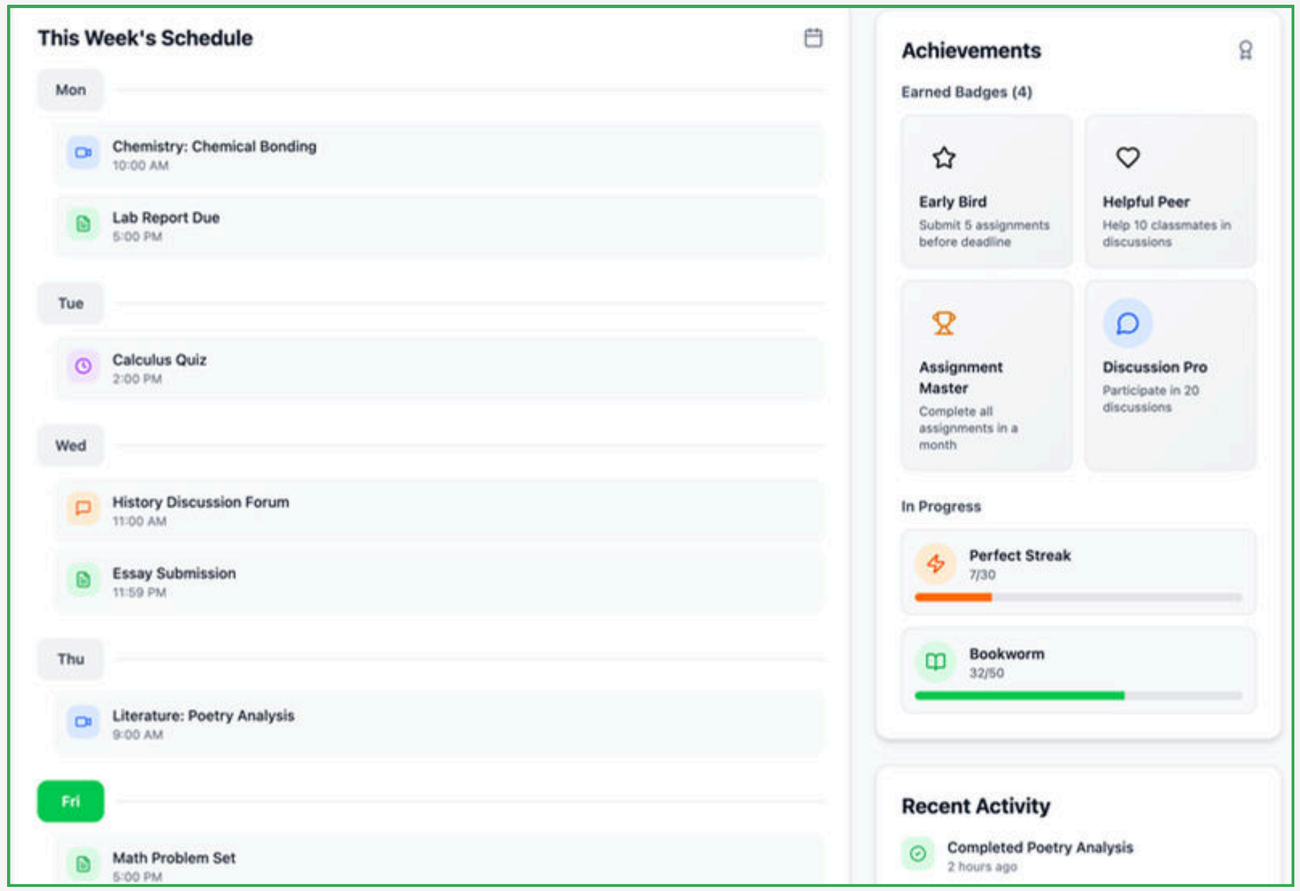
Dashboard - Student

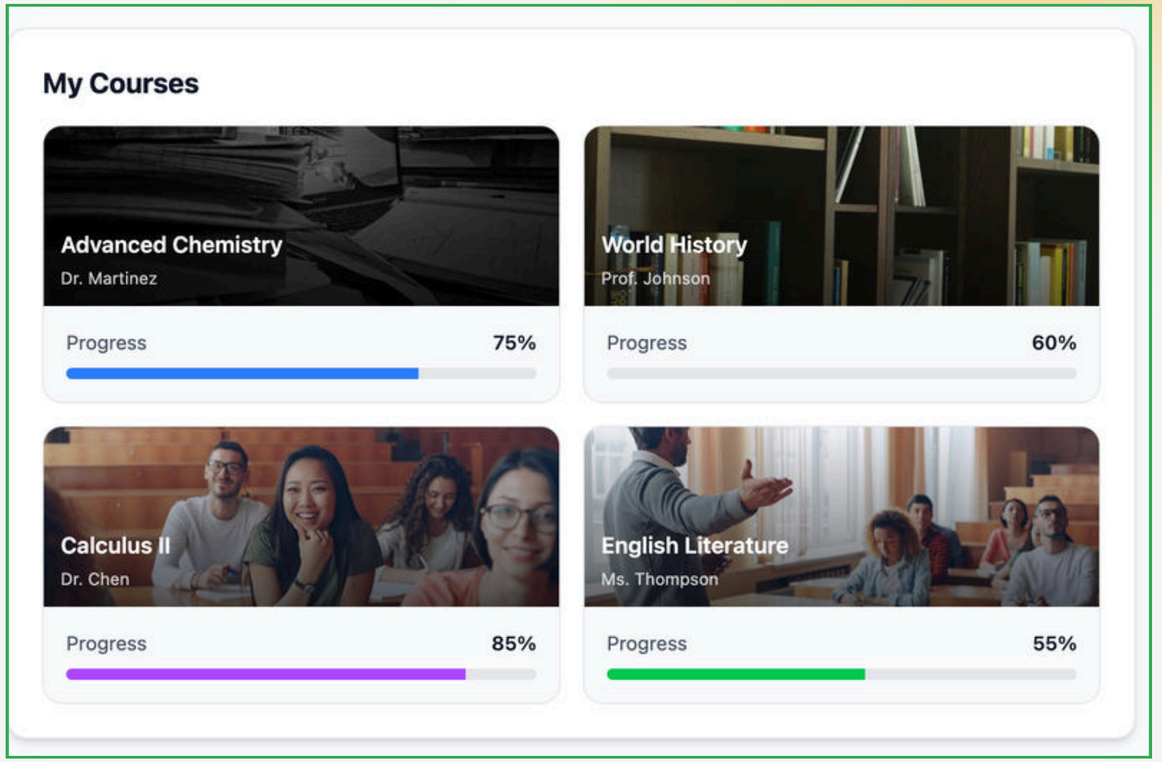
Video of Prototype

AI Assistant



Redesign

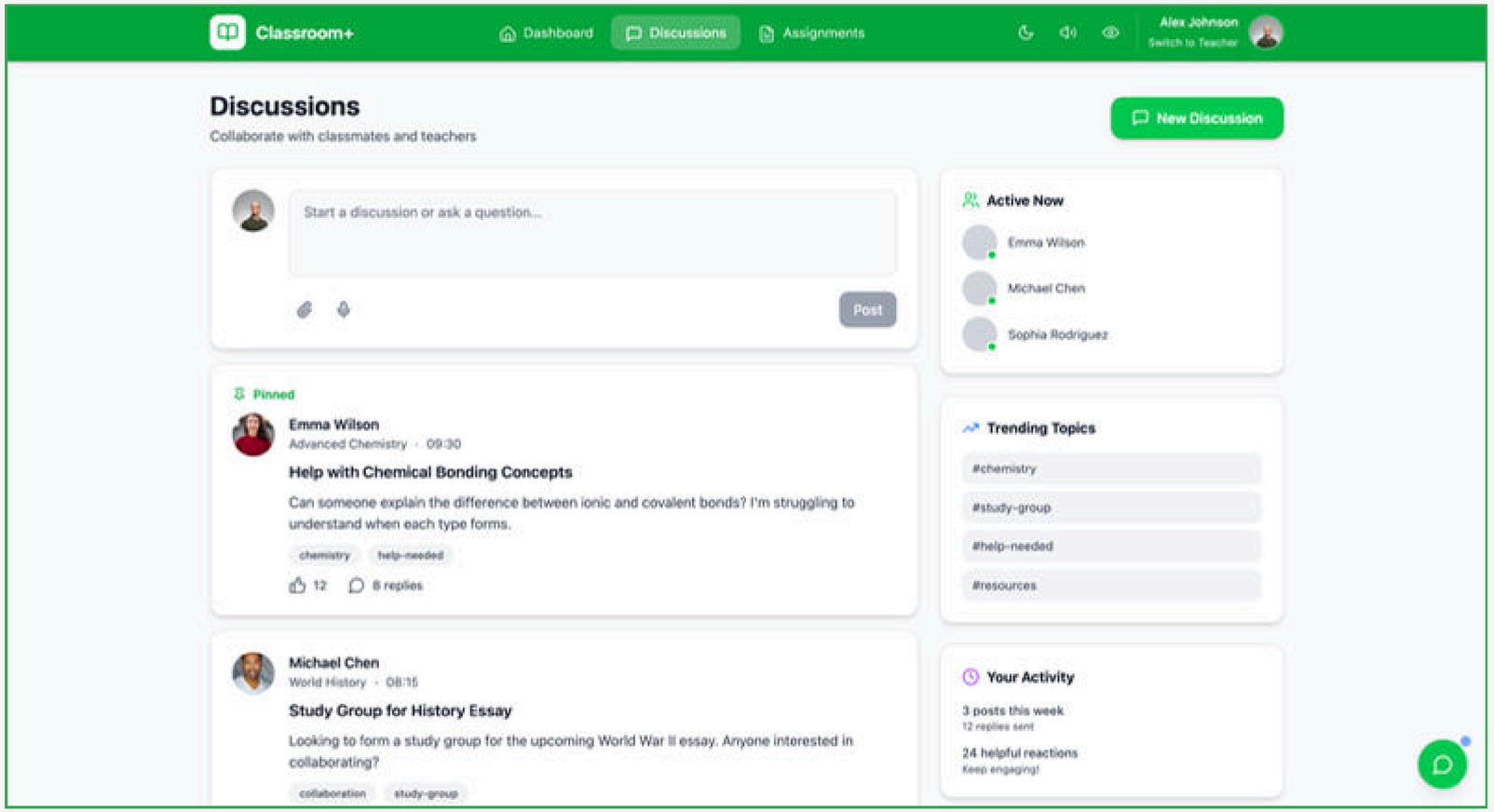




Activity Tracker - Student

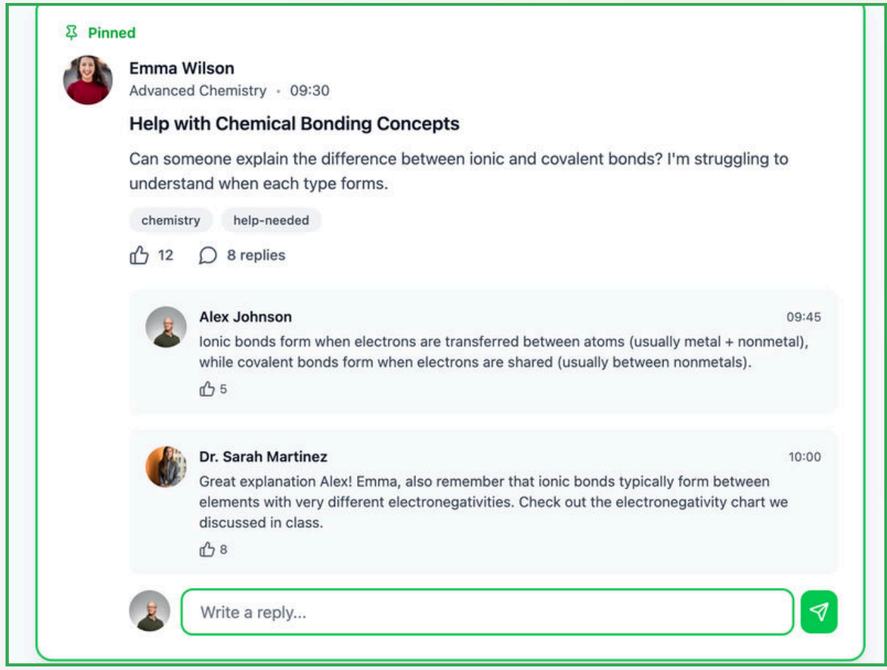
Video of Prototype

Redesign



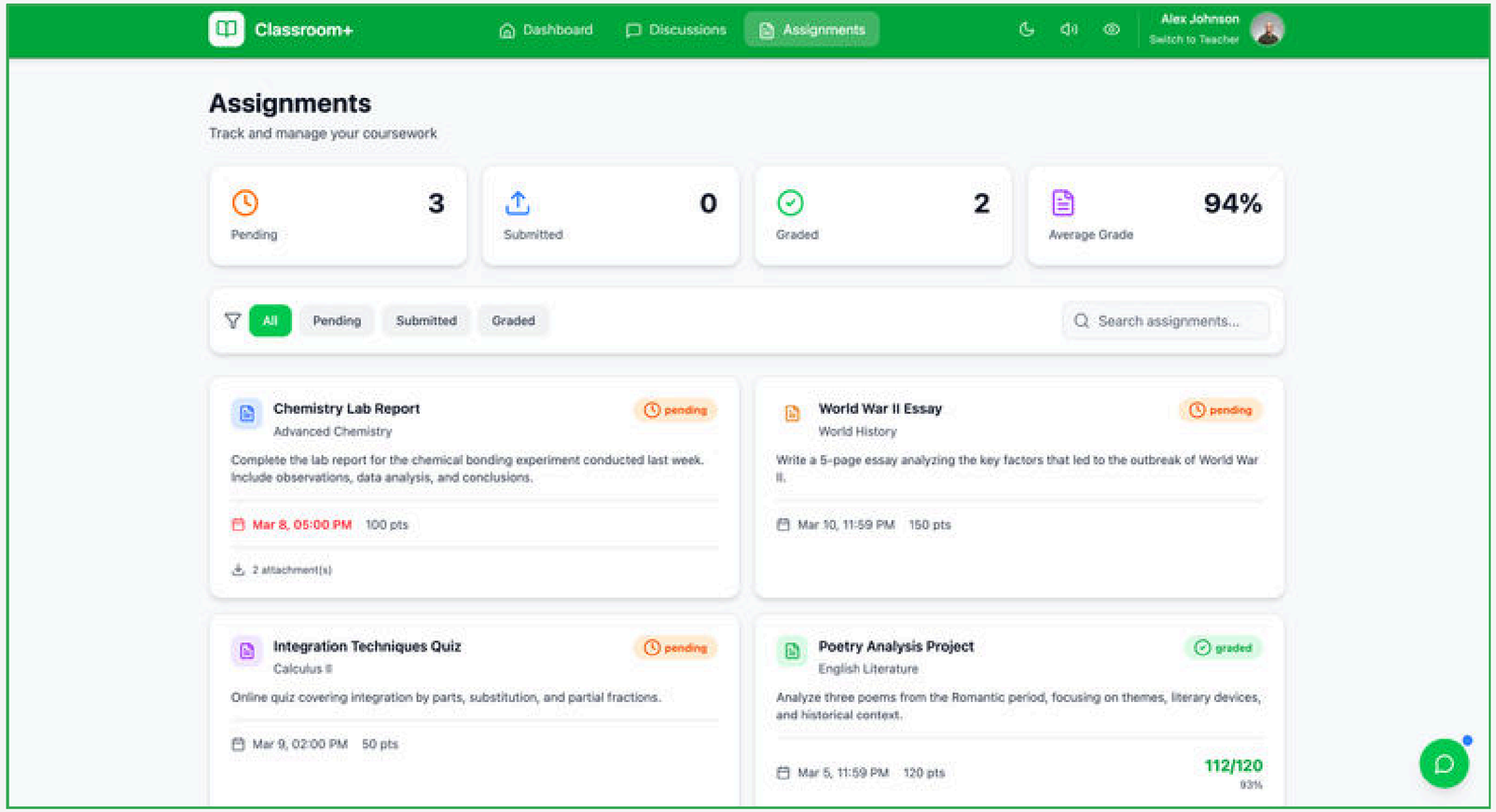
Dicussion Window

Easy reply



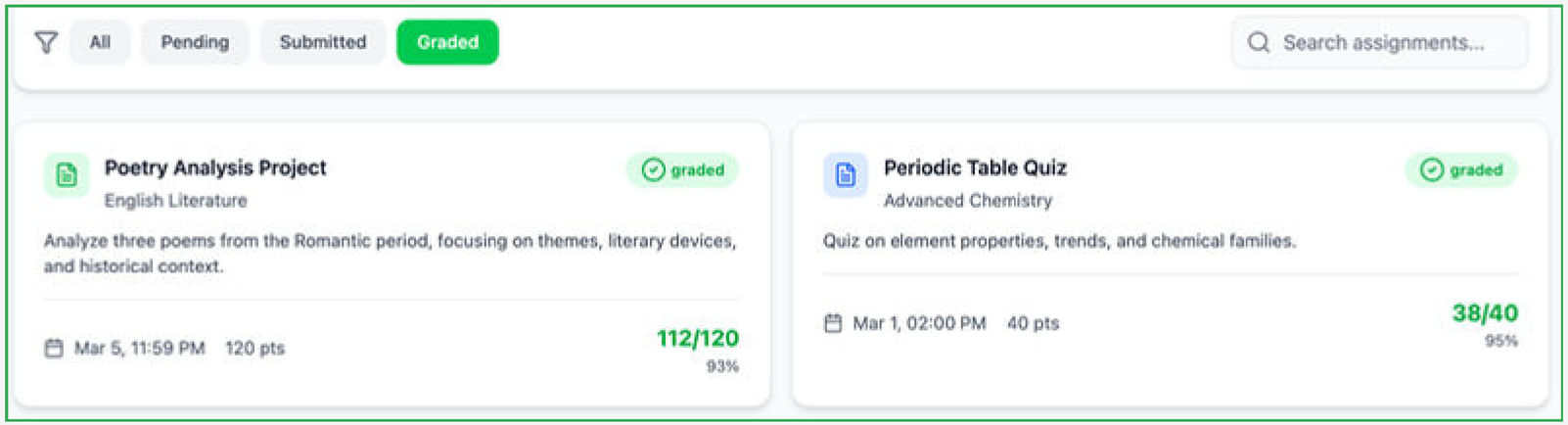
Video of Prototype

Redesign



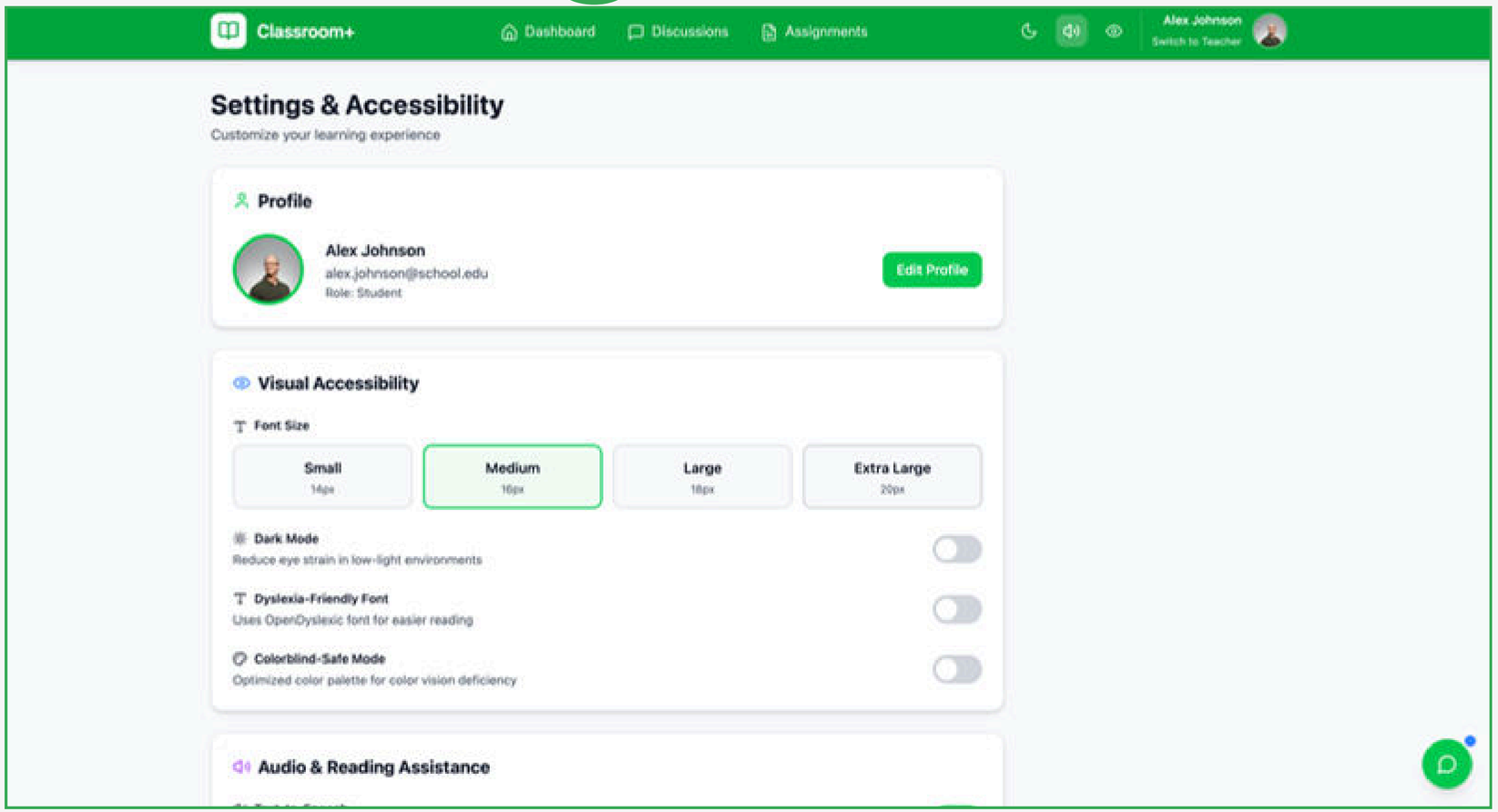
Activity Depending on the Assignment Status

Assignment Window



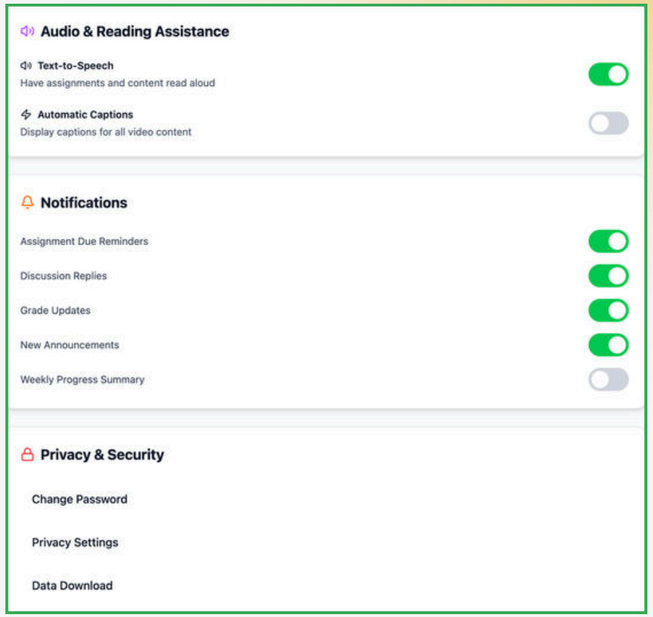
Video of Prototype

Redesign



Profile window

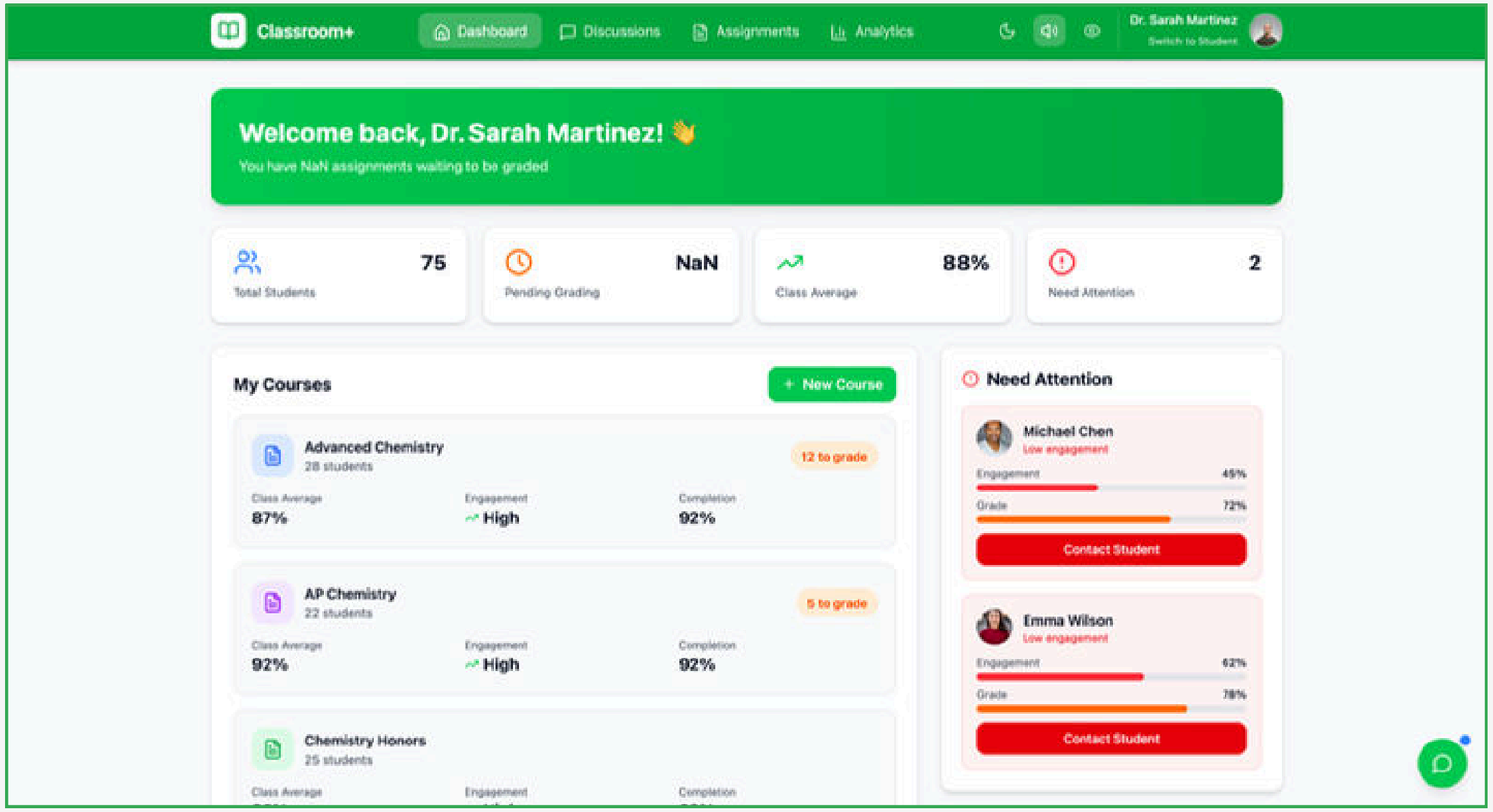
Privacy & Customisation





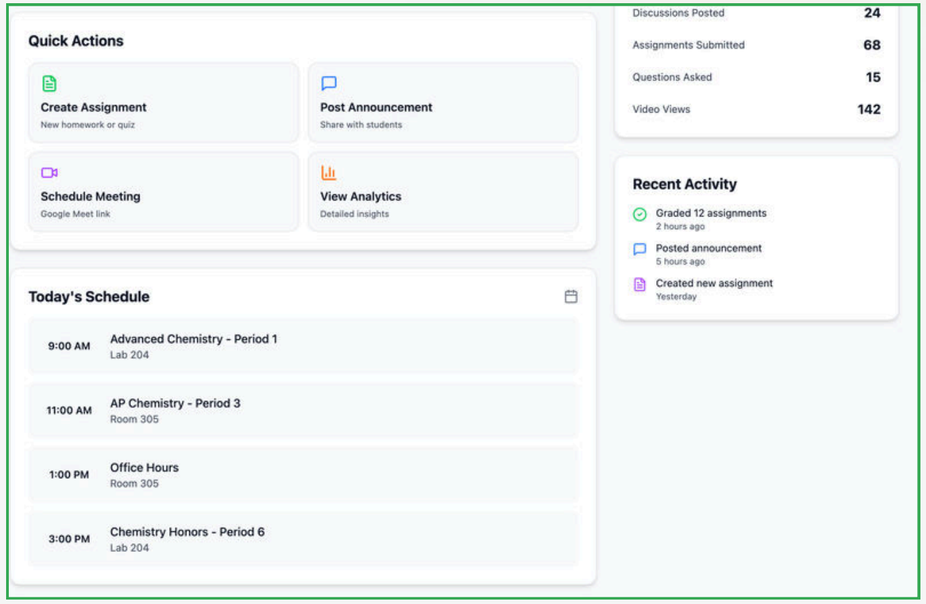
Video of Prototype

Redesign



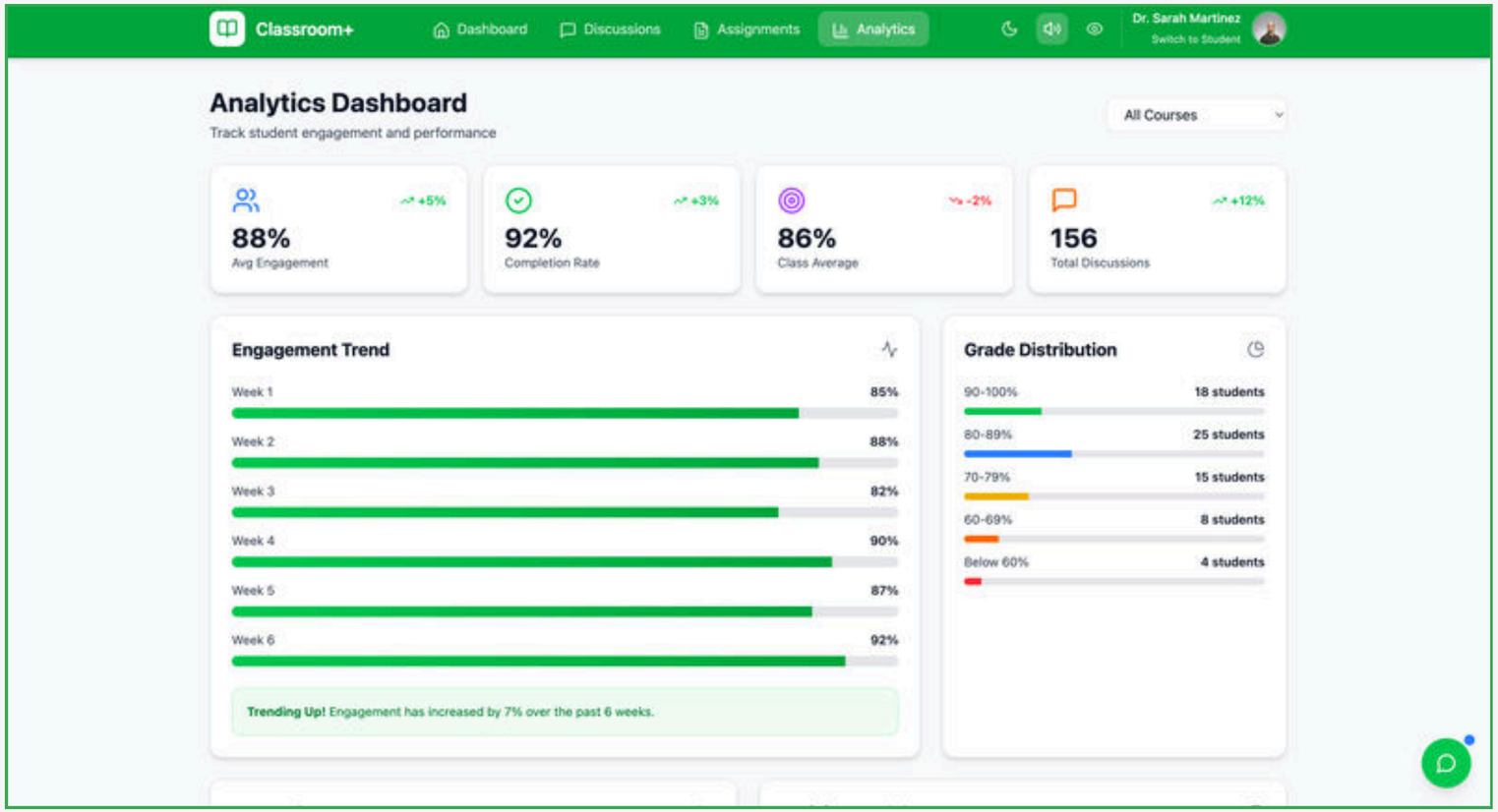
Dashboard - Teacher

Activity Tracker - Teacher



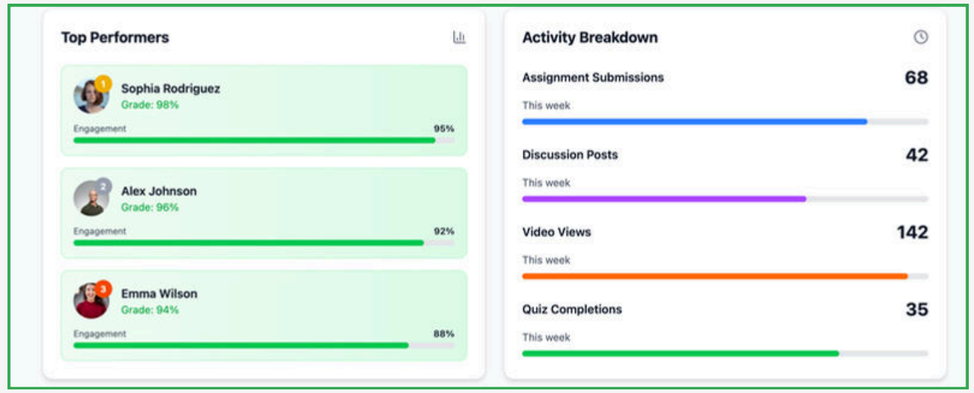
Video of Prototype

Redesign



Analytics - Teacher

Student Tracker - Teacher

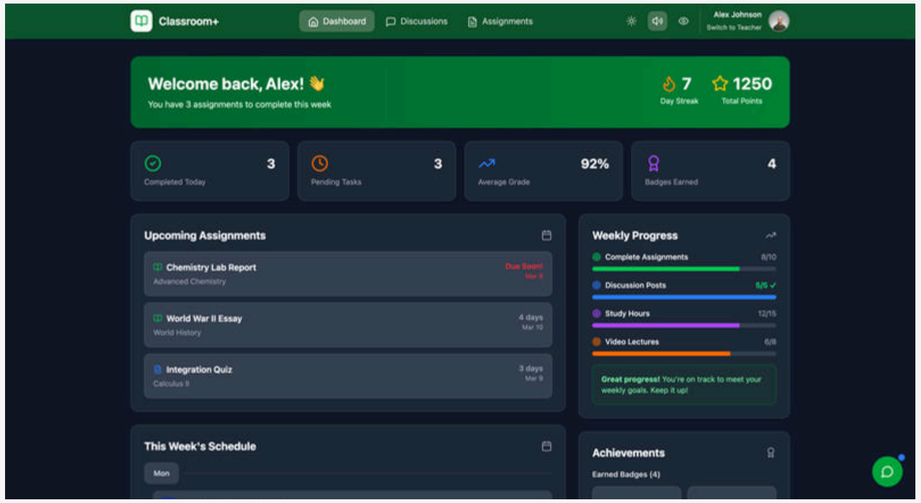


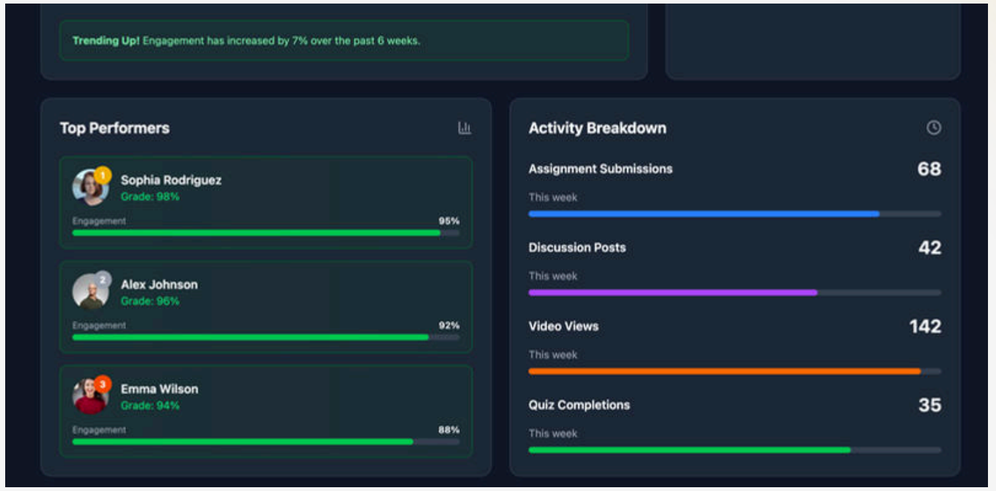
Video of Prototype

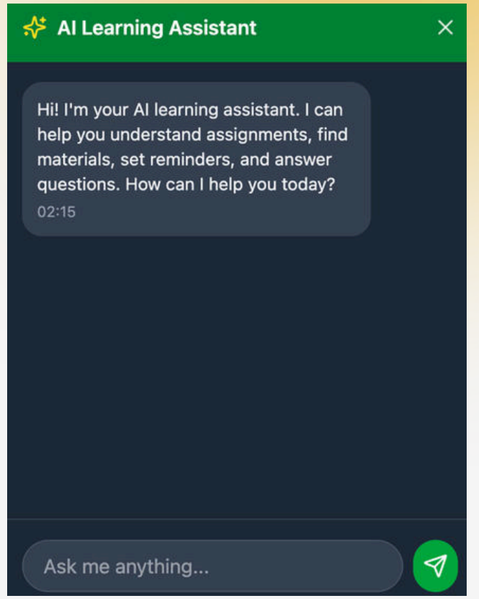
[Video of Prototype]

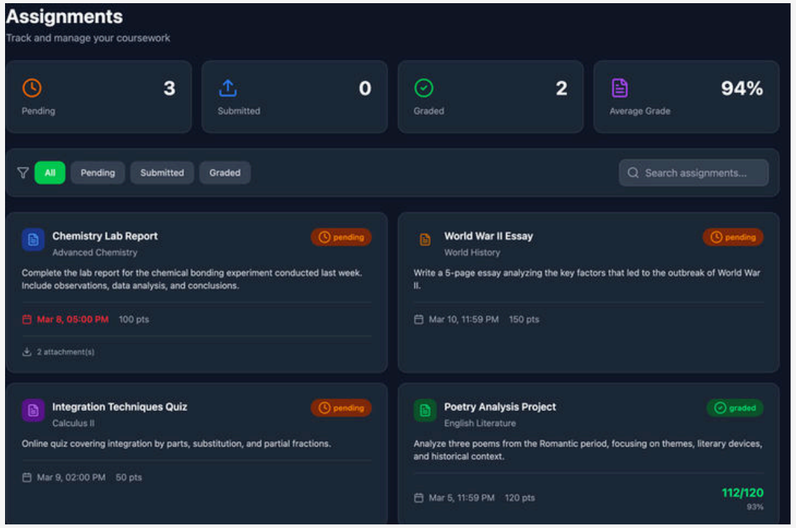
Redesign

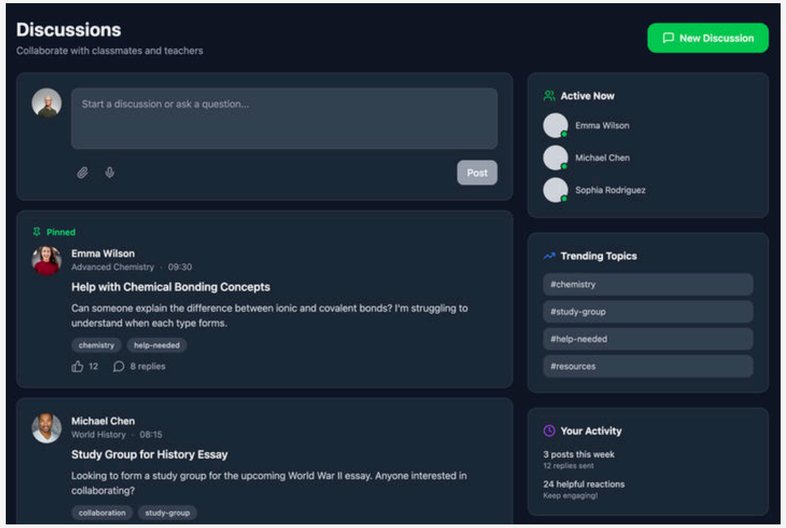
Additional Dark mode feature

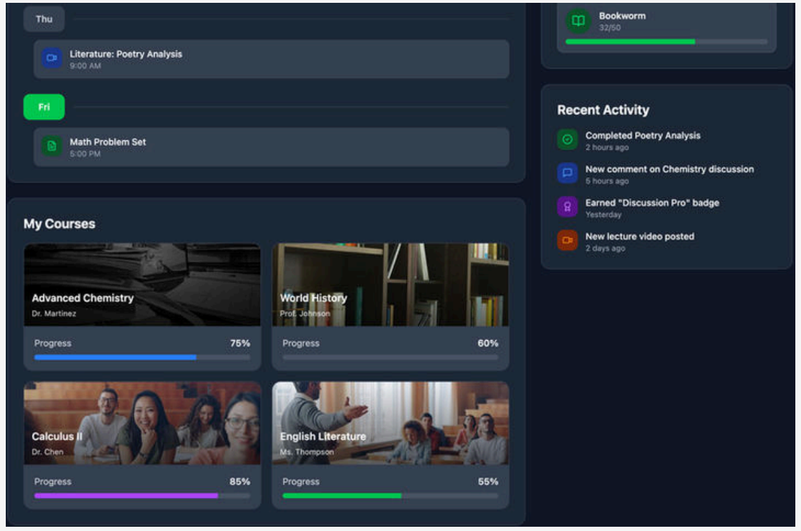












Original vs Redesigned Platform

Original Google Classroom	Redesigned Version
Basic interface focused mainly on assignment posting and submission	Interactive dashboard that provides a complete overview of assignments, schedule, and activity
Assignments appear in a simple list, making it difficult to track priorities	Assignments are clearly categorized into **upcoming, completed, and overdue** for better task management
Limited motivation or engagement features	**Gamification system** with progress indicators and rewards to encourage participation
No integrated academic support tools	**AI Assistant** helps students understand concepts and get help with assignments
Minimal visibility of recent classroom updates	**Recent Activity section** shows announcements, graded work, and important updates
No structured view of schedules	**Schedule view** allows students to manage classes and deadlines easily
Teachers have limited insights into student engagement	**Teacher Activity Dashboard** provides analytics on student participation and performance
Mostly static learning environment	**Interactive and student-centered learning experience**

Impact of the Redesign

The redesigned version of Google Classroom enhances the overall digital learning experience by improving engagement, organization, and learning support for both students and teachers.

Improved Student Engagement
The introduction of gamification elements such as progress indicators and rewards encourages students to stay motivated and actively participate in coursework.

Better Assignment Management
Clear assignment categorization and a centralized student dashboard help learners easily track upcoming, completed, and overdue tasks, reducing confusion and improving time management.

Enhanced Learning Support
The integrated AI assistant provides students with instant academic guidance, helping them understand concepts and complete assignments more effectively.

Improved Information Accessibility
Features like the recent activity feed and schedule view ensure students stay updated with announcements, deadlines, and class events in a structured manner.

Data-Driven Teaching Insights
The teacher activity dashboard allows educators to monitor student participation and performance, helping them identify struggling students and provide timely support.

Overall Outcome
The redesign transforms the platform from a basic classroom management tool into a more interactive, organized, and supportive digital learning environment.



Thank You!

Video of Prototype

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